



CS683: Advanced Computer Architecture

Lecture 15: It is the Memory Stupid! 😊

<https://www.cse.iitb.ac.in/~biswa/courses/>

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DRAM Organization

Channel

DIMM

Rank

Chip

Bank

Row

Column

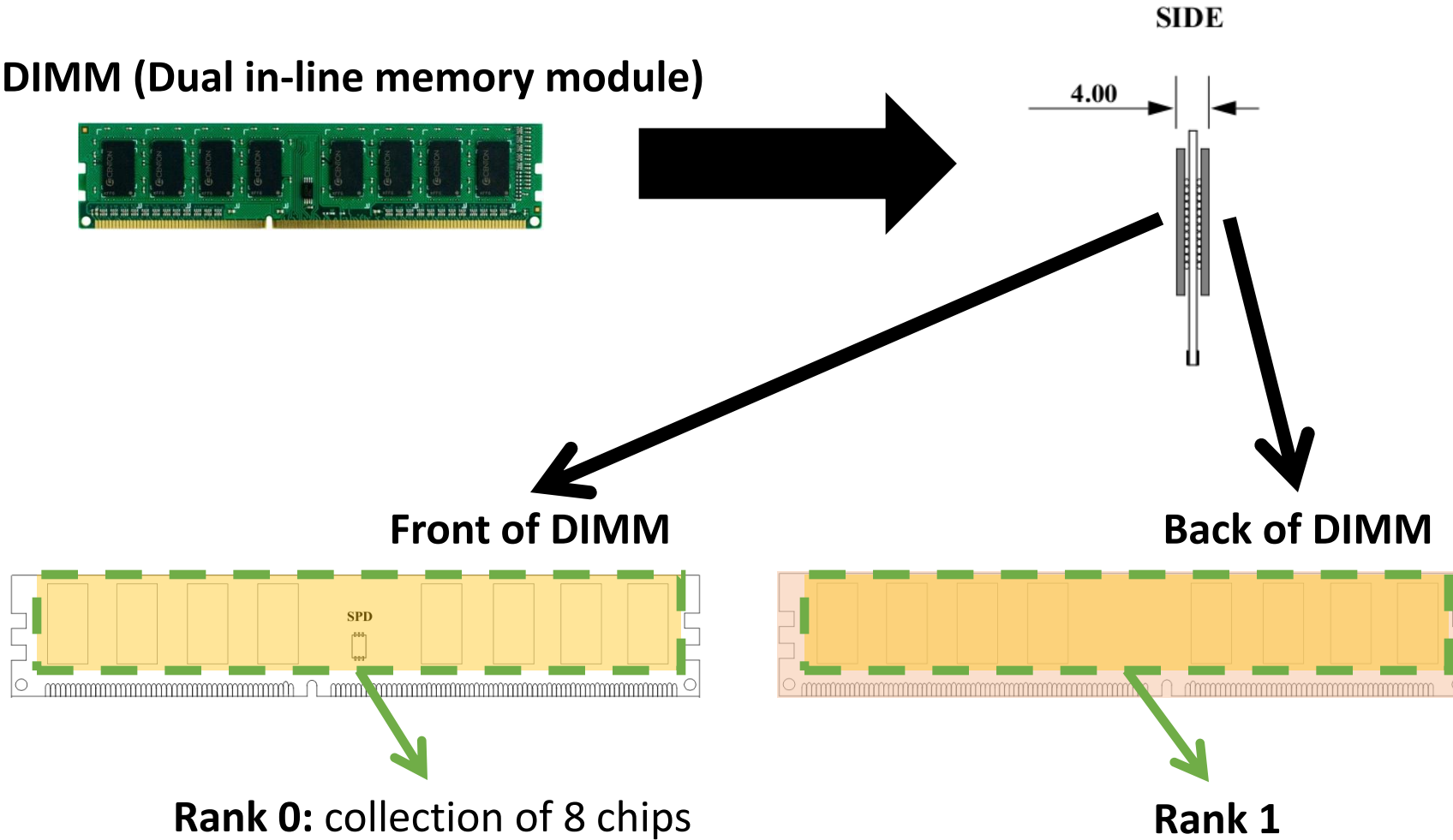


Rank 1 with 8 chips

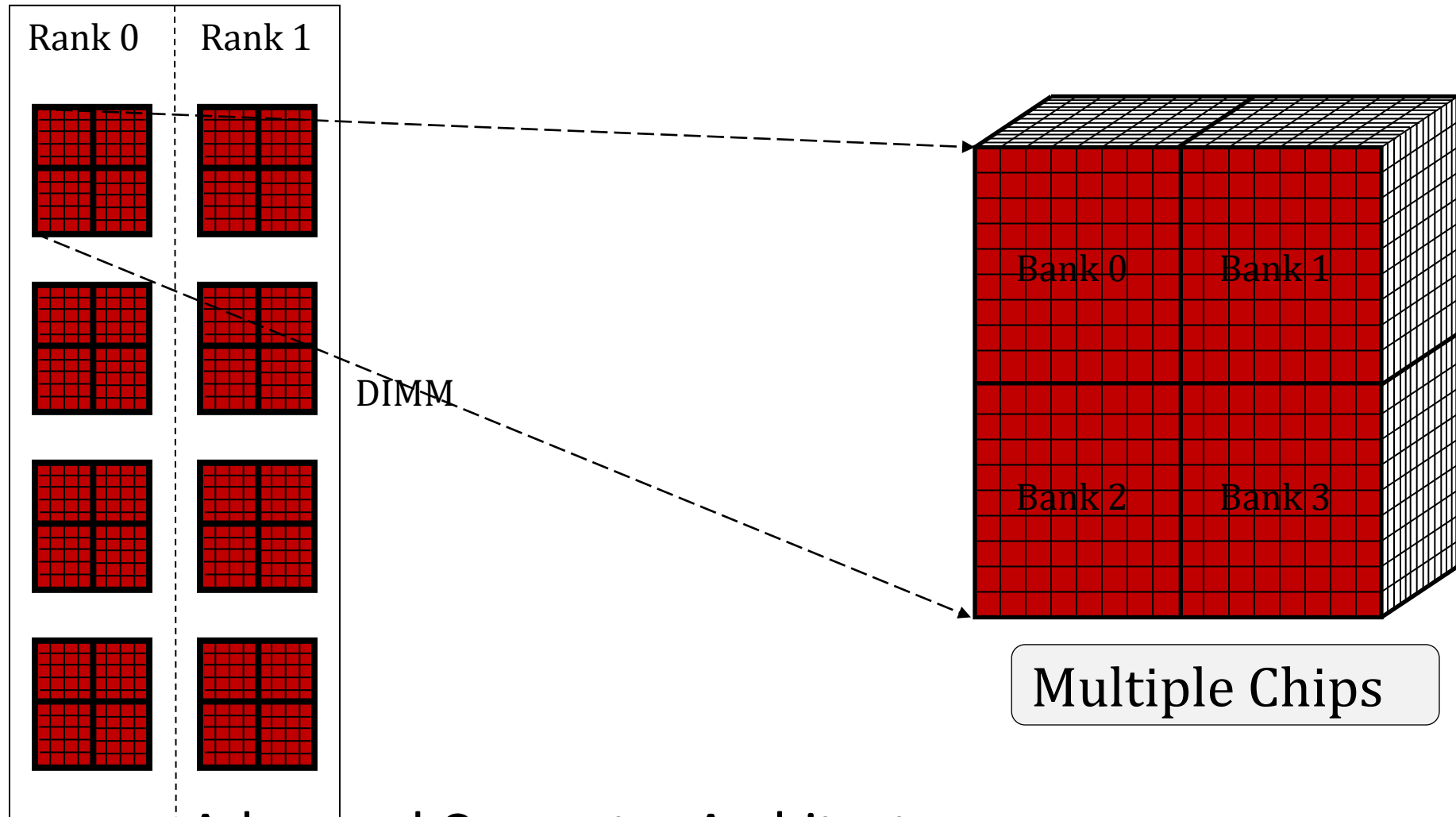
Rank 0 with 8 chips

Rank

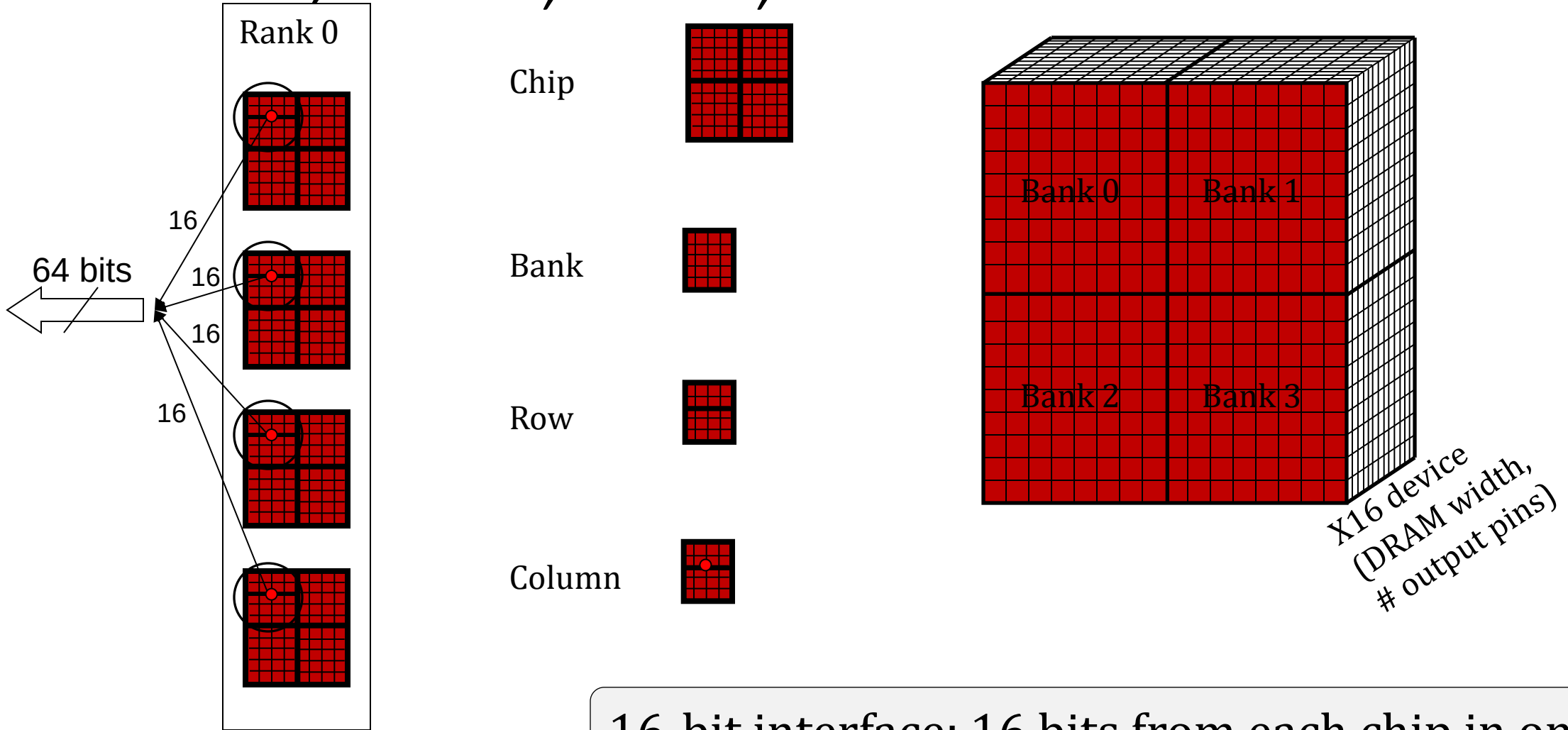
DIMM (Dual in-line memory module)



Ranks, Banks, Rows, Columns

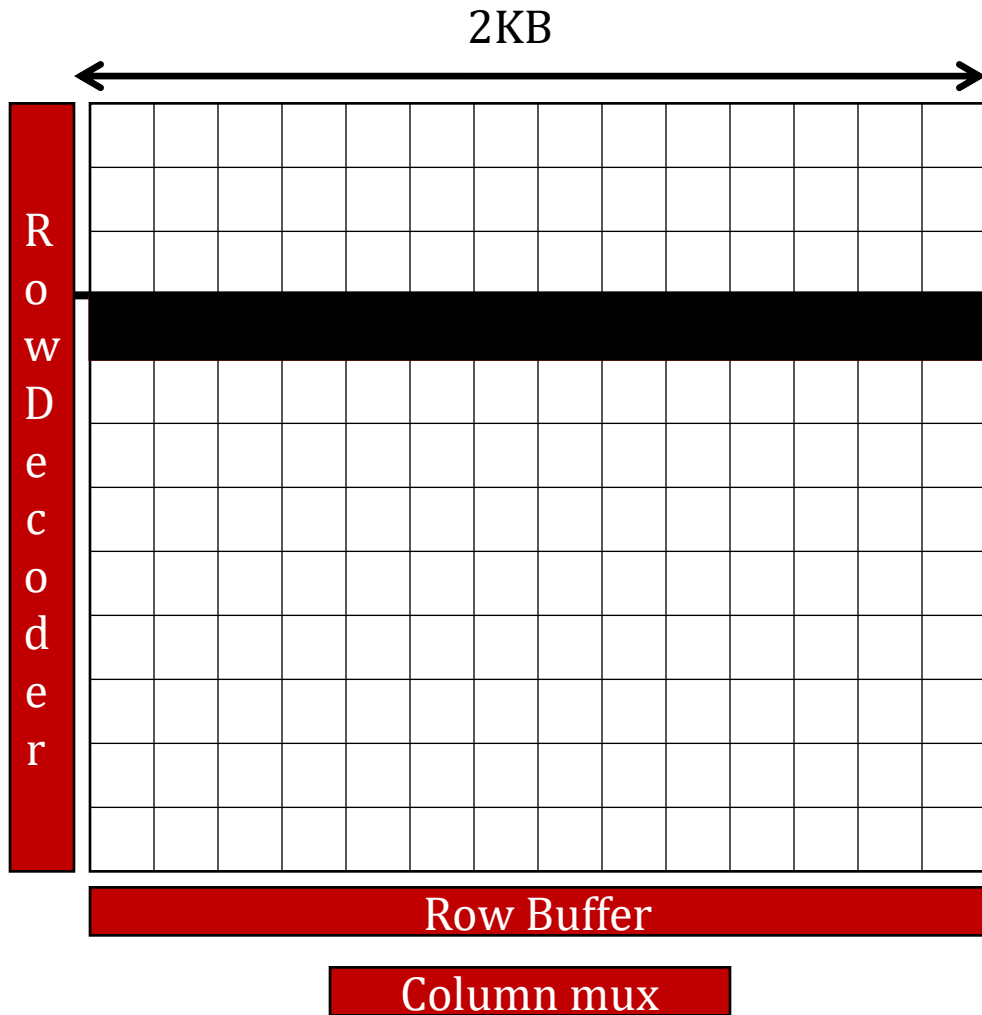


Ranks, Banks, Rows, Columns



16-bit interface: 16 bits from each chip in one go

Row Decoder and Row buffer



Each bank has a row buffer

Stores the last used row

An Example

2Gb * 8 DRAM Chips (one side of the rank)

Total 16 chips + 2 chips for ECC (for both the ranks)

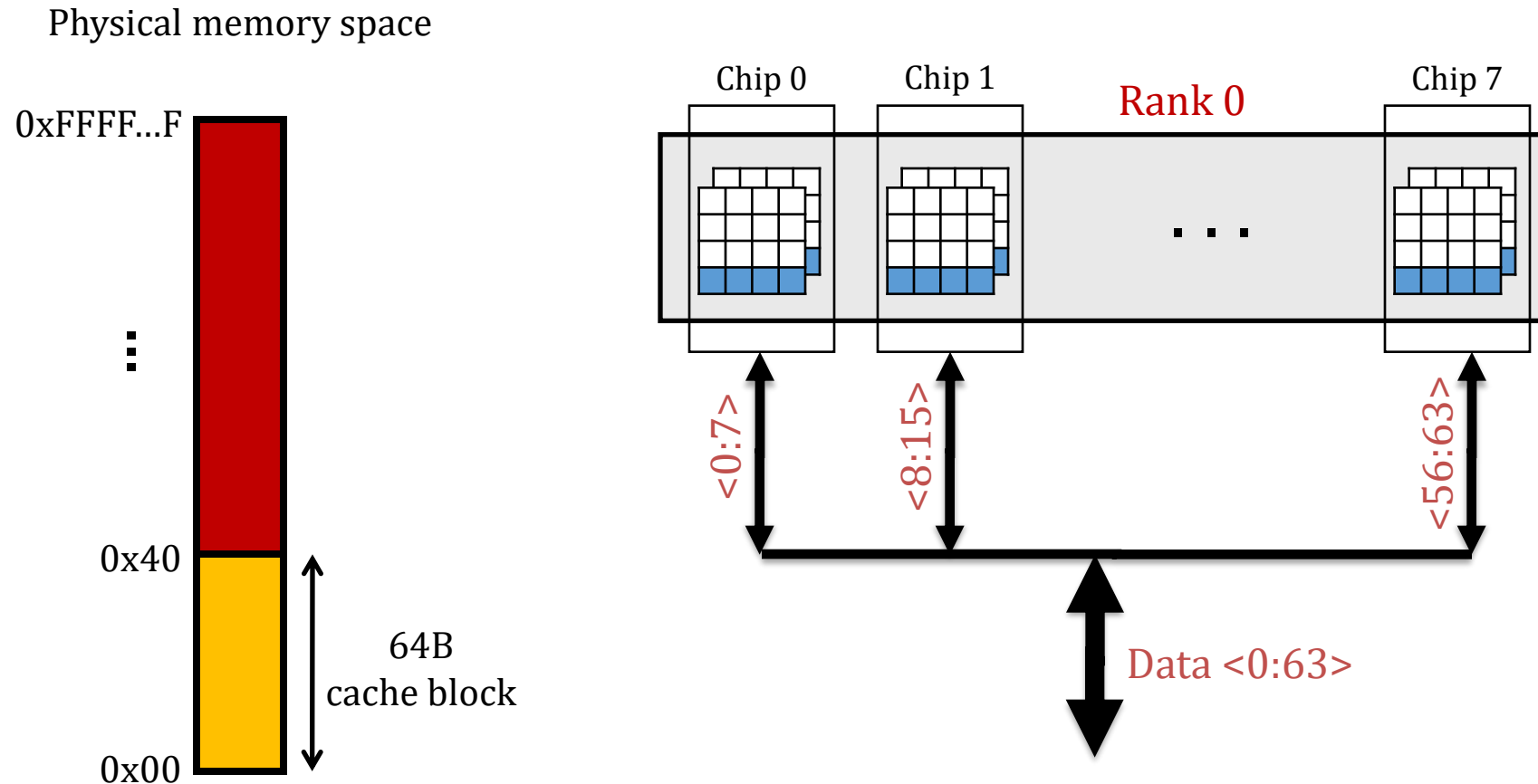
64 bit + 8 bit ECC interface (72 bit wide DIMM)

Transferring a 64B cache line will take 8 transfers of 8B each

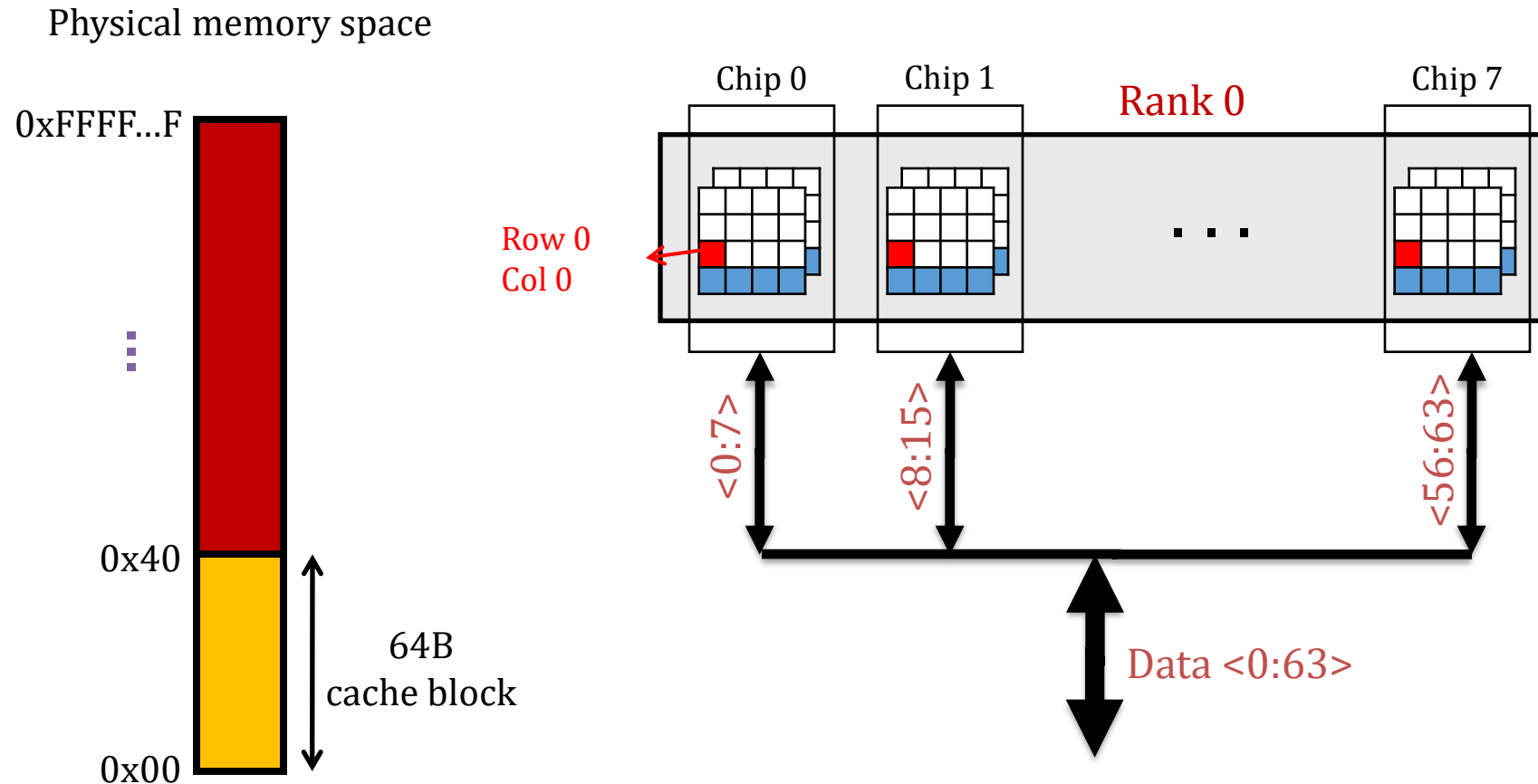
8B will come from 8 chips (8 bits from one chip)

1 bit from each DRAM array assuming 8 DRAM arrays per bank

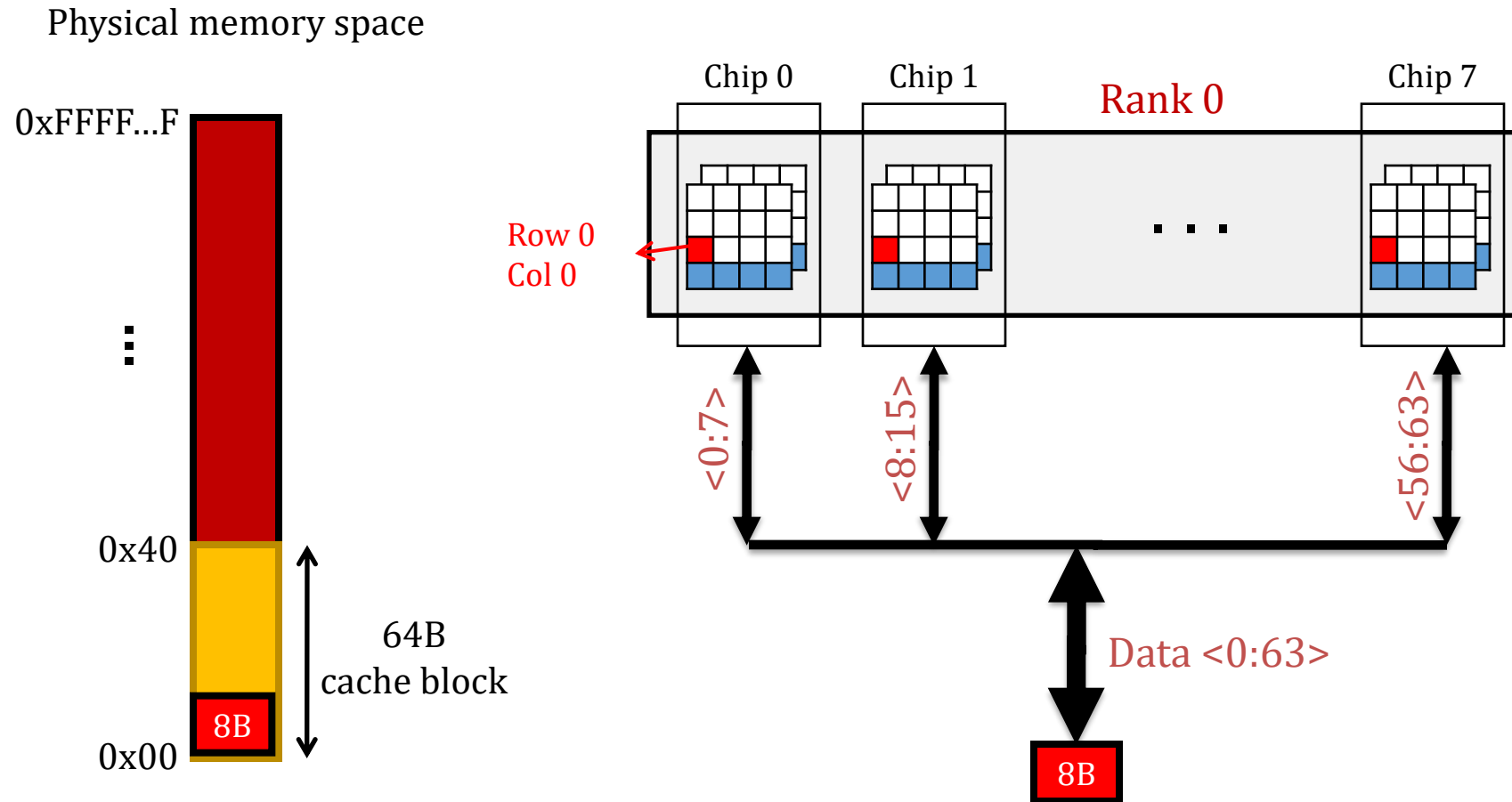
DRAM to LLC interaction



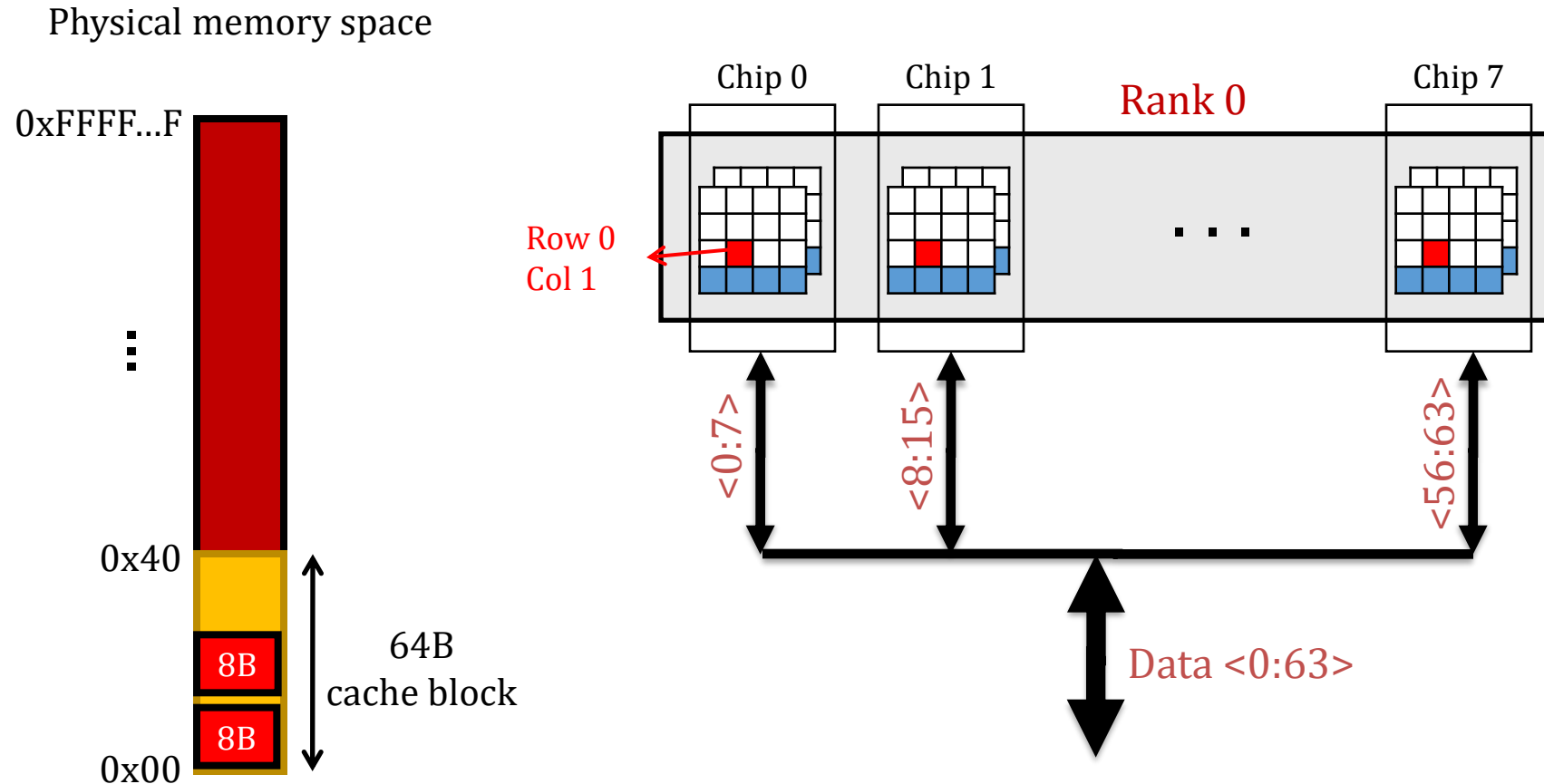
DRAM to LLC interaction



DRAM to LLC interaction



DRAM to LLC

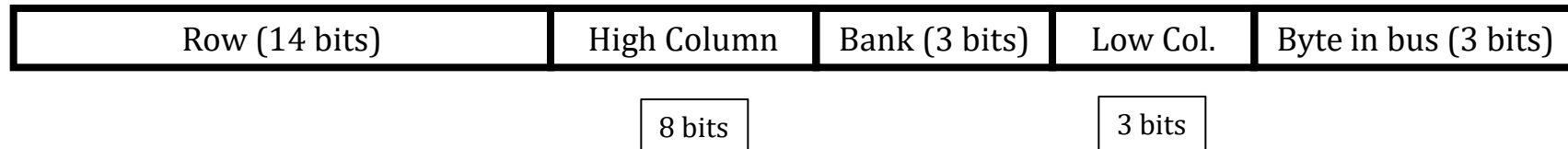


8 cycles (DRAM IO): 1 cycle transfers 8 bytes from a column

DRAM Address Mapping (One Channel)

2GB DRAM, 8 Banks, 16K rows, 2K Columns per bank

Cache Interleaving: Consecutive cache blocks in consecutive banks



Row Interleaving: Consecutive rows in consecutive banks



Row Access

A “DRAM row” is also called a “DRAM page”

Each address is a <row,column> pair

Access to a “closed row”

- **Activate** command opens row (placed into row buffer)
- **Read/write** command reads/writes column in the row buffer
- **Precharge** command closes the row and prepares the bank for next access

Access to an “open row”

- No need for activate command

Row Buffer hit/miss/conflict

Access Address:

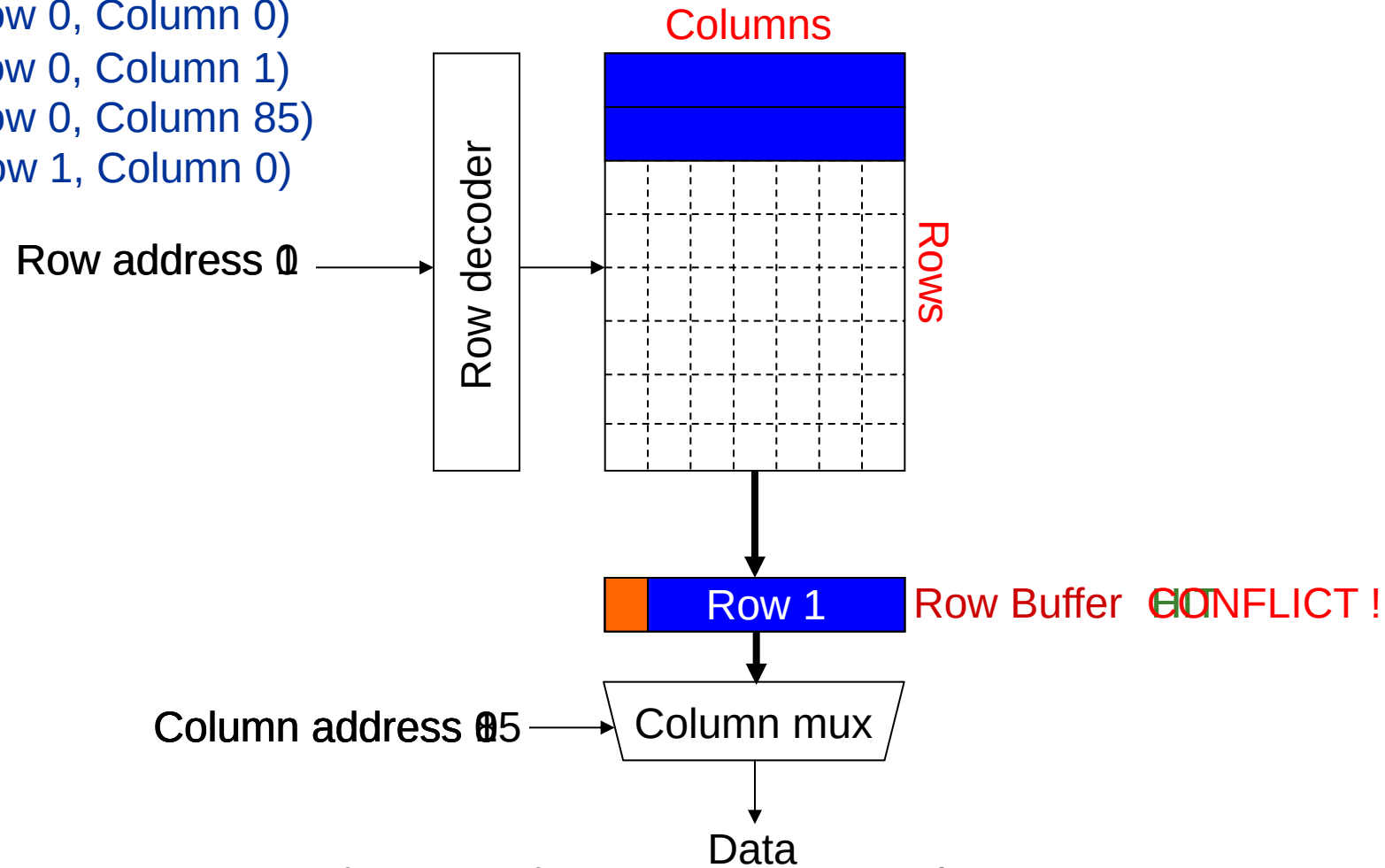
(Row 0, Column 0)

(Row 0, Column 1)

(Row 0, Column 85)

(Row 1, Column 0)

Row address 0



DRAM Timing Constraints

tRAS: Row address strobe latency

tRP: Precharge latency

tCAS: Column address strobe latency

tRC: Row cycle time: $tRAS + tRP$

Row (Page) Policies

Open Page (do not get confused with an OS page)

After an access: Keep the **page in the row-buffer**

Consecutive accesses to the same page : **Row-buffer Hit**

On an access to different page: **Close the row and open the new one**

Closed page:

After an access: Close the page if there are no accesses to the same row in the request queue.

Consecutive accesses to different page : **Low latency**

On an access to different page: **No need to close the row**

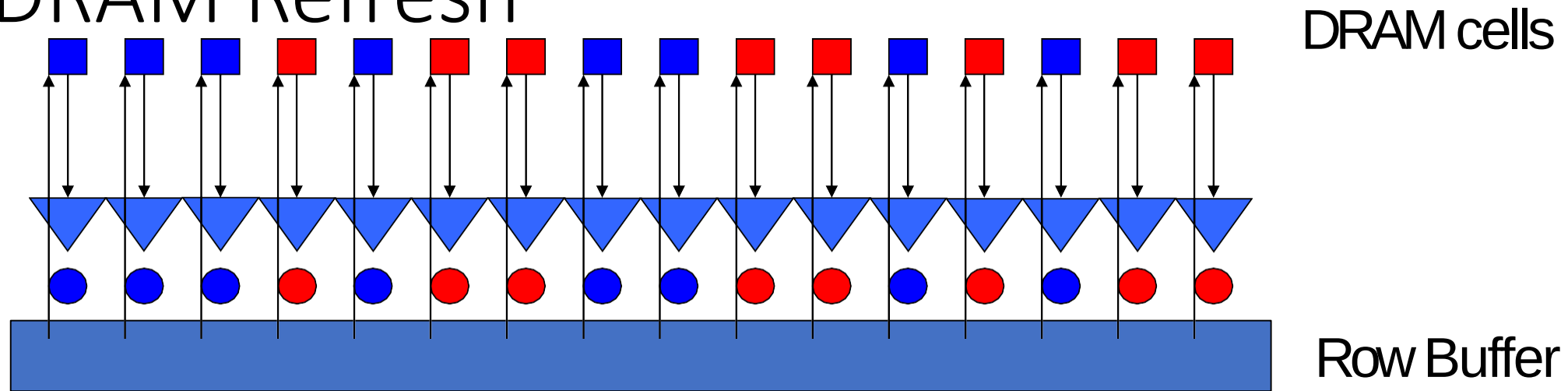
10K view on the latency

Page Empty: ACT + CAS

Page Hit: CAS

Page Miss: PRE+ACT+CAS

DRAM Refresh



DRAM cells lose contents after awhile,
Refresh command refreshes all rows at different granularity

How to implement a refresh? What is the latency? How frequent?

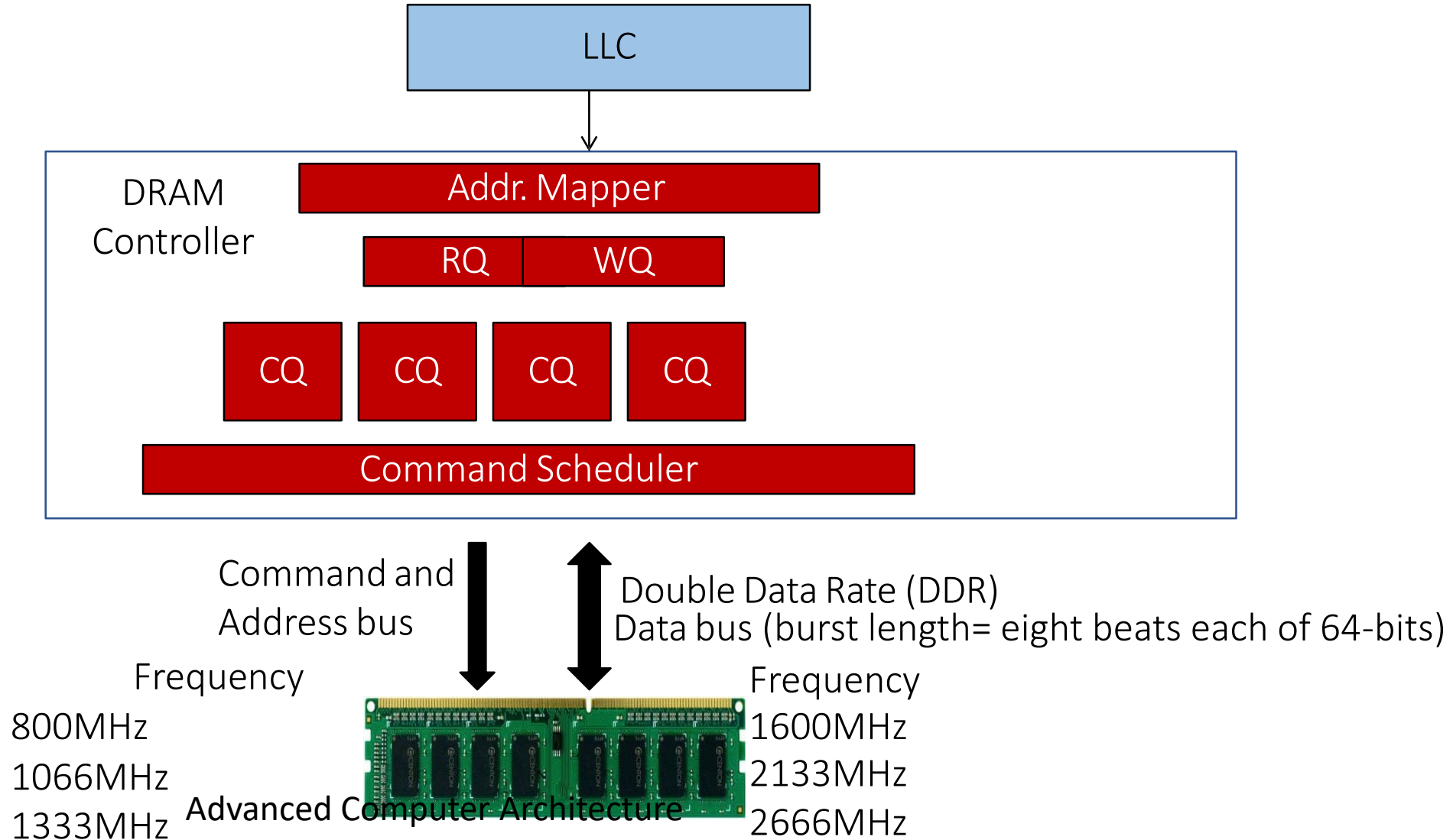
DRAM cells are refreshed every 64 ms at normal temperature ($< 85^{\circ}\text{C}$).



PAUSE



DRAM Controller

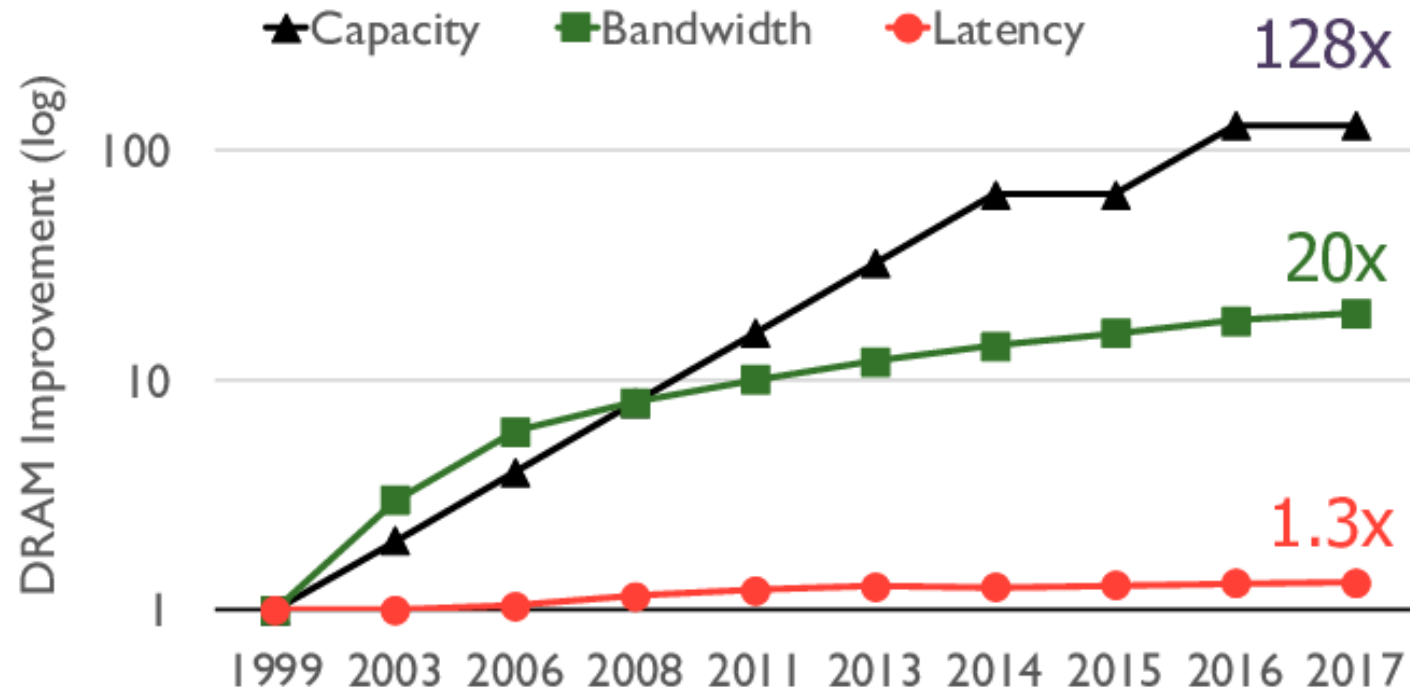


DRAM Bandwidth

Names	Memory clock	I/O bus clock	<u>Transfer rate</u>	Theoretical bandwidth
DDR-200, PC-1600	100 MHz	100 MHz	200 MT/s	1.6 GB/s
DDR-400, PC-3200	200 MHz	200 MHz	400 MT/s	3.2 GB/s
DDR2-800, PC2-6400	200 MHz	400 MHz	800 MT/s	6.4 GB/s
DDR3-1600, PC3-12800	200 MHz	800 MHz	1600 MT/s	12.8 GB/s
DDR4-2400, PC4-19200	300 MHz	1200 MHz	2400 MT/s	19.2 GB/s
DDR4-3200, PC4-25600	400 MHz	1600 MHz	3200 MT/s	25.6 GB/s
DDR5-4800, PC5-38400	300 MHz	2400 MHz	4800 MT/s	38.4 GB/s
DDR5-6400, PC5-51200	400 MHz	3200 MHz	6400 MT/s	51.2 GB/s

Memory Trends: Bandwidth, capacity, and others

DRAM Capacity, Bandwidth & Latency



Need more bandwidth? High Bandwidth Memory (HBM) & The world of DRAM Cache

- Base logic layer (blue), CPU and Memory glued by TSV
- Multiple DRAM chip layers
- Significantly higher bandwidth (200 to 800 GB/s)
- What is the problem then? Capacity ☹️
- HBM3: 819.2 GB/sec **Capacity: 24GB** ☹️

